Technical Architecture and Feature Enhancement Strategy for cricketcoach.online

The global sports technology sector has transitioned from manual-assist review tools and basic overlay graphics into a highly automated domain powered by real-time computer vision, edge-native deep learning, and multi-view spatial reconstruction.[1] For an online platform such as cricketcoach.online to deliver competitive value to academies, professional coaches, and individual athletes, its underlying software architecture must transcend basic video playback.[3] It must evolve into an interactive, multi-angle biomechanical feedback and ballistic trajectory projection system.[5]

To guide the development team in upgrading cricketcoach.online, this report outlines the necessary mathematical foundations, system architectures, and algorithmic pipelines. These upgrades will enable the platform to track joint keypoints, project ball trajectories, synchronize multi-angle cameras, and support scalable club administration.[3]

Competitive Benchmarking and Feature Translation

To establish a defensible market position, the enhanced architecture for cricketcoach.online must bridge the gap between human-centric communication systems and fully automated biomechanical engines.[4] Existing digital coaching systems are highly fragmented; some excel at league statistical aggregation but lack automated computer vision pipelines [9], while others provide precise biomechanical tracking but are constrained by single-sport applications, proprietary hardware, or high system latency.[5]

The table below provides an architectural comparison of prominent sports technology platforms and open-source implementations. This benchmarking highlights the technical gaps that cricketcoach.online must address to establish a competitive advantage.

Platform	Core Technical Modality	Multi-Angle Sync Strategy	Biomechanical Target and Resolution	System Architecture Bottlenecks	Business Model & Pricing
Ludimos [3]	2D Ball, Bat, and Player Tracking via Cloud AI [3]	Real-time platform synchronization [11]	Joint keypoint detection and kinogram generation	High upload latency; documented system instability [11]	Tiered SaaS; target-focused B2B Academy

			[11]		AMS
PitchVision [13]	Multi-Sensor Fusion and Local Video Cataloging [14]	4-Camera Synchronized Hardware Arrays [14]	Foot position tracking on the crease and release height [14]	High dependency on local laptop processing and proprietary sensors [14]	Freemium mobile application with hardware monetization [13]
CricHeroes [9]	Match Scoring and Statistical WebInsights [9]	None	None (Manual input for ball-by-ball events) [9]	Absent computer vision or automated video analysis features [9]	Free basic scoring; premium SaaS for leagues and associations [9]
SportsReflector [5]	2D/3D Pose Scoring and AR Overlay [5]	Integrated Multi-Angle Video Processing [5]	25+ Joint tracking with injury risk alert system [5]	Lack of sport-specific (cricket-specific) contextual rules [4]	$19.99/mo (Pro) to $49.99/mo (Coach) [4]
SwingVision [4]	Ball Trajectory and Court Boundary Triangulation [4]	None	Ball velocity and court impact locations [4]	Biomechanical pose tracking of the athlete is not supported [4]	$29.99/mo consumer subscription [4]
HomeCourt [4]	Real-time Basketball Shot and Dribble Tracking [4]	None	Basketball-specific trajectory arc and release mechanics [4]	Restricted to court-based ball mechanics [4]	Freemium model starting at $7.99/mo [4]

Cricket Coach AI (Srinath Sankara) [15]	Offline 2D Pose Estimation via MediaPipe Lite [15]	None	2D joint-angle threshold checking [15]	Local single-thread execution; Flask-bound processing [15]	Open-source reference implementation [15]

By building upon these models, cricketcoach.online can stand out by offering high-precision biomechanics, real-time ball tracking, and multi-angle synchronization within a web-based, hardware-agnostic platform.[6]

Biomechanical Pose Estimation and Rule Engines

The core of the technical upgrade involves converting raw video frames into precise, actionable joint kinematics.[6] Rather than delivering non-personalized scoring metrics, the platform should implement a dedicated, rule-based biomechanical engine.[15] This engine evaluates joint coordinates against established physiological safety and performance parameters.[15]

Biomechanical Keypoint Tracking Mechanics

Using a monocular or multi-view setup, the pipeline tracks a standard human skeleton using model nodes mapped via MediaPipe or YOLO Pose.[8] Joint angles θ are calculated dynamically using the dot product of two directional vectors $\mathbf{u}$ and $\mathbf{v}$ derived from coordinate triplets:

$$\mathbf{u} = \mathbf{P}_{\text{joint\_1}} - \mathbf{P}_{\text{joint\_vertex}}$$

$$\mathbf{v} = \mathbf{P}_{\text{joint\_2}} - \mathbf{P}_{\text{joint\_vertex}}$$

$$\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right) \times \left(\frac{180}{\pi}\right)$$

This mathematical approach supports the real-time detection of complex movement patterns.[15] The table below lists the primary biomechanical checks, their spatial thresholds, and the target drills provided to the athlete.

Biomechanical Metric	Target Vector Mapping (Landmarks)	Expected Angle Range (θ)	Physiological or Performance Pathology	Targeted Remedial Drill

Bowler Knee Landing [15]	Hip-Knee-Ankle (Front Foot)	$155^\circ -$	Collapsing below 155° transfers force directly to the lumbar spine, inducing stress fractures.[15]	**Braced-Leg Landing Drill:** Repetitive delivery stride onto a firm, extended front knee focusing on hip-shoulder separation.
Bowler Elbow Extension [6]	Shoulder-Elbow-Wrist (Bowling Arm)	$\Delta\theta \leq$ during delivery	Exceeding a 15° straightening threshold classifies the action as an illegal delivery ("chucking").[6]	**Wall Release Drill:** Stationary bowling action with the throwing arm sliding against a vertical boundary to enforce a straight arm path.
Bowler Guide Arm Drive [15]	Shoulder-Elbow-Wrist (Non-Bowling Arm)	$\geq$	Insufficient guide arm extension limits trunk rotation torque and reduces ball release velocity.[15]	**High-Reach Target Drill:** Reaching to touch an overhead marker with the non-bowling hand during the gathering phase of the action.
Bowler Shoulder Turn [15]	Mid-Shoulder to Mid-Hip Plane	$\geq$ (Relative to Crease)	Insufficient hip-to-shoulder separation (front-on delivery)	**Side-on Alignment Drill:** Bowling alongside a linear guide

			increases torsional lumbar strain.[15]	while maintaining the hips parallel to the pitch direction until front-foot contact.
Batter Knee Flex [15]	Hip-Knee-Ankle (Front Leg at Impact)	$150^\circ -$	A fully locked knee prevents rapid weight transfer; a collapsed knee destroys shot control and balance.[15]	**Low-Drive Holding Drill:** Executing front-foot drives and maintaining the post-shot stance for 3 seconds to measure muscle activation.
Batter Head Alignment [15]	Inter-Ear Vector to Gravity Axis	$0^\circ \pm$ deviation	Head tilting alters binocular visual depth tracking, causing premature bat closing and leg-before-wicket risks.[15]	**Eye-Level Drop Ball:** Executing drives against vertically dropped balls while keeping the eyes aligned horizontally.

Injury Risk Classification and Safety Models

Beyond performance metrics, the platform should integrate automated joint safety modeling.[5] Biomechanical errors often correlate with specific physiological injuries, which can be predicted and flagged by tracking joint loading patterns over time.[8]

- **Lumbar Spine Strain Detection**: This is evaluated by tracking the alignment between the hip and shoulder axes during the release phase.[15] Bowlers who display a mixed delivery action (combining side-on hips with a front-on chest rotation) exhibit higher torsional lumbar strain.[15] The platform flags this risk when the relative angle between the shoulder plane and hip plane drops below 40° during front-foot contact.

- **Knee Joint Valgus Instability**: In both batters and bowlers, sudden lateral knee displacement under load presents a significant ACL injury risk.[8] The system monitors the lateral knee deviation vector, flagging a warning if the knee joint collapses inward relative to the vertical ankle-to-hip alignment axis during the loading phase.[8]
- **Shoulder Impingement Hazards**: The platform evaluates the rotation angle of the shoulder joint during ball release.[8] Release heights that fall below head height (characteristic of "round-arm" actions) place high shear stress on the rotator cuff.[15] The system measures the vertical offset of the wrist relative to the head landmark to identify and flag these actions.[15]

Adaptive Thresholding Architecture

A generic, one-size-fits-all approach is insufficient for professional athletic coaching.[15] The rule engine must feature an adaptive configuration framework that automatically shifts evaluation thresholds based on two primary player attributes:

- **Age-Group and Skeletal Maturity**: Physical tolerances vary across age groups due to differences in joint flexibility and bone density.[15] The platform dynamically adjusts its threshold profiles for four separate brackets: Under 10, Under 15, Under 18, and Adult.[15] For instance, a front-knee collapse threshold that triggers a warning at 155° for an adult is adjusted to a wider tolerance of 148° for an Under-10 player to reflect natural physical development.[15]
- **Skeletal Handedness Mirroring**: To ensure seamless, plug-and-play video uploading, the platform must dynamically handle left-handed and right-handed athletes.[15] Upon player registration, the system automatically configures coordinate mapping: for right-handed batters, the front knee is mapped to the left joint landmark, whereas for left-handed batters, it mirrors to the right joint landmark, ensuring precise calculations across all views.[15]

Multi-View Spatial Reconstruction and Temporal Alignment

To address the request for "more angles" on cricketcoach.online, the technical team must implement synchronized, multi-camera video playback.[5] This allows coaches and players to analyze a single athletic movement from multiple angles simultaneously (e.g., side-on, front-on, and umpire views).[14]

Software-Based Temporal Synchronization

Hardware-based genlock (which links cameras via physical cables) is impractical for everyday coaching environments.[22] Consequently, the platform must utilize a software-based temporal synchronization pipeline.[7]

The system utilizes two primary synchronization techniques:

- **Visual-Acoustic Impact Alignment**: The software automatically parses the video streams to detect transient physical events.[1] It runs parallel detection loops: a visual loop that identifies peak changes in joint and ball velocity [23], and an acoustic loop that detects the high-frequency audio signatures of ball-to-bat contact or ball-to-pitch impact.[1] The frame matching this transient event is set as the global zero index (t_0), and all other video feeds are temporally shifted to align with this point.[7]
- **VisualSync Epipolar Optimization**: When tracking unposed and unsynchronized handheld camera feeds in the field, the platform uses epipolar constraints.[7] The algorithm identifies co-visible moving 3D points (such as the ball or joint landmarks) across different camera feeds.[7] By calculating and minimizing the epipolar error across these coordinate sequences, the system determines the temporal latency offset between cameras down to a sub-50 millisecond resolution without requiring manual input.[7]

Camera Calibration and Multi-View Triangulation

To construct an accurate 3D pose, the system requires camera calibration.[18] This process determines the internal geometry of each camera (intrinsic matrix $\mathbf{K}$) and its physical position in space (extrinsic matrix $$).[18]

The projection mapping between a 3D world point $\mathbf{X}$ and its 2D camera coordinate $\mathbf{x}$ is defined as:

$$\mathbf{x} \sim \mathbf{P}\mathbf{X} = \mathbf{K}\mathbf{X}$$

Once the projection matrices $\mathbf{P}_1$ and $\mathbf{P}_2$ are computed for at least two overlapping camera views, a linear triangulation method is applied via OpenCV's cv2.triangulatePoints.[18] This converts synchronized 2D keypoints into consistent, metric 3D coordinates [18]:

$$\mathbf{A}\mathbf{X} = \mathbf{0}$$

$$\mathbf{A} = \begin{bmatrix} x\mathbf{p}^{3T} - \mathbf{p}^{1T} \\ y\mathbf{p}^{3T} - \mathbf{p}^{2T} \\ x'\mathbf{p}'^{3T} - \mathbf{p}'^{1T} \\ y'\mathbf{p}'^{3T} - \mathbf{p}'^{2T} \end{bmatrix}$$

This 3D reconstruction resolves structural occlusions (e.g., when the body blocks a camera's line of sight), maintaining tracking continuity and providing high-precision biomechanical insight.[25]

Synced Front-End Video Comparison Dashboard

For the user interface, the front-end must support side-by-side video comparisons.[5] Using optimized WebGL rendering and libraries like DualView, the interface allows players to compare their mechanics with past deliveries or professional players.[27]
The table below outlines the core requirements for the synchronized front-end player component on cricketcoach.online.

Technical Feature	Engineering Target & Library	User Interaction & Visual Presentation
Frame-Accurate Synchronization [28]	Canvas rendering utilizing requestAnimationFrame [15]	Scrubbing one video timeline automatically updates the paired video frame to maintain exact temporal alignment.[28]
Variable Slow-Motion Controls [13]	HTML5 Video element rates: $0.25\times, 0.5\times, 1.0\times$ [13]	Users can toggle down to $0.25\times$ speed, triggering slow-motion visual tracking trails.[15]
Dynamic A-B Looping [28]	Custom Javascript state tracking with I/O indices [28]	Allows the user to select custom start and end points to loop specific phases of movement, such as the delivery stride or backswing.[28]
Side-by-Side to Overlay Swipe [28]	WebGL fragment shaders and slider controls [28]	A horizontal slider allows users to transition smoothly from side-by-side viewing to a semi-transparent video overlay for direct form comparison.[28]
GPU-Accelerated Export [28]	MediaRecorder API compiling WebGL canvas output [28]	Encodes the side-by-side comparison with synchronized drawings and text overlays into an MP4 file for sharing.[28]

Ball Tracking and 3D Trajectory Projection Engines

To deliver professional-grade analytics, the platform must track the physics of the delivery.[6] Ball tracking is technically difficult because high ball speeds can cause motion blur, resulting in frame detection gaps.[23]

Object Detection and State Interpolation

The recommended architecture runs a dual-model pipeline in parallel.[6] The player is isolated using specialized instance segmentation (such as SAM2 or YOLOv8 Pose), while the ball path is tracked using custom-trained, heatmap-based deep convolutional networks like TrackNet or YOLOv11.[6] When tracking objects as small as a cricket ball, processing should run on high-speed footage (60–120 FPS) to maintain a continuous, high-fidelity visual history.[6]

To resolve frames where the ball is occluded by the bat, body, or background clutter, the tracking engine employs a constant velocity Kalman filter combined with Piecewise Cubic Hermite Interpolating Polynomials (PCHIP).[23]

The state vector $\mathbf{x}_k$ of the ball at frame k is represented as:

$$\mathbf{x}_k = \begin{bmatrix} x & y & v_x & v_y \end{bmatrix}^T$$

$$\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k$$

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k$$

Where $\mathbf{F}$ is the state transition matrix, $\mathbf{H}$ is the observation matrix, and $\mathbf{w}_k, \mathbf{v}_k$ denote the Gaussian process and measurement noise matrices respectively.[32] When the observation engine fails to register a target detection (i.e., $\mathbf{z}_k$ is null), the PCHIP interpolation algorithm reconstructs the missing coordinates by enforcing continuous first derivatives along the ball's curved trajectory:

$$h_i = x_{i+1} - x_i$$

$$d_i = \frac{y_{i+1} - y_i}{h_i}$$

The derivative s_i at node i is calculated locally to prevent overshoot, maintaining physical consistency around the bounce point [23]:

$$s_i = \begin{cases} \frac{3d_{i-1}d_i}{(2d_i+d_{i-1})+(d_{i-1}+2d_i)} & \text{if } \text{sign}(d_{i-1}) = \text{sign}(d_i) \\ 0 & \text{otherwise} \end{cases}$$

This mathematical smoothing allows the platform to automatically detect the bounce event.[33] It marks the peak deceleration and immediate directional inversion of v_y to isolate the pitch map landing zone.[32]

Depth Estimation and Physics-Based Calibration

To convert raw 2D pixel tracks into a metric 3D representation without expensive stereo-camera configurations, the model calibrates the visual space against a standard physical reference: the 22-yard (20.12 m) pitch distance.[6] By feeding the known distance boundaries and camera focal properties into a monocular depth estimation pipeline, the platform maps the relative pixel-to-metric scaling factor [6]:

$$s = \frac{D_{\text{pitch\_meters}}}{D_{\text{pitch\_pixels}}}$$

Once scaled, a 3D trajectory reconstruction model projects the post-bounce path forward.[6] This enables the application to predict the height and angle at which the ball passes the batter, creating a DRS-style ball-tracking projection directly from standard smartphone footage.[6]

Ballistics and Trajectory Metrics

By measuring the reconstructed ball flight path, the processing engine extracts key trajectory metrics [6]:

$$\mathbf{v}_{\text{release}} = \frac{\|\mathbf{P}_1 - \mathbf{P}_0\|}{\Delta t}$$

$$\mathbf{a} = \frac{\mathbf{v}_k - \mathbf{v}_{k-1}}{\Delta t}$$

This physical calculation produces the core ballistic metrics shown in the table below.

Ballistic Parameter	**Measurement Formula / Core Algorithm**	**Tactical Significance & Coaching Application**
Release Velocity [6]	First derivative of position vector at boundary entry [6]	Measures raw athletic power, tracking variations in ball velocity across a

		bowling spell.[14]
Post-Bounce Speed [6]	First derivative of position vector immediately following v_y inversion [6]	Evaluates pitch friction and decay, showing how much speed the ball loses after bouncing.
Line & Length Mapping [6]	2D pixel intersection mapped to 3D pitch coordinate space [6]	Automatically aggregates deliveries into pitch map visual charts.[6]
Swing/Spin Deviation [6]	Transverse displacement offset relative to linear projection [6]	Quantifies ball movement in the air or off the pitch to measure delivery quality.[6]
Vertical Bounce Profile [14]	Maximum vertical peak coordinate post-bounce [14]	Measures delivery bounce height to help batters analyze their response to short-pitched bowling.[14]

Front-End Interactive UX and Analytical Overlay Systems

The technical success of cricketcoach.online depends on delivering complex biomechanical data through an intuitive, interactive user interface.[15] Rather than presenting static data charts, the front-end must render dynamic visual overlays that highlight movement patterns in real time.[15]

Interactive Canvas Overlay Mechanics

The front-end rendering engine uses the HTML5 Canvas API paired with WebGL to overlay tracking graphics on top of native video feeds.[15]

- **Pulsing Focus Rings**: When a joint movement error is identified, the system renders a pulsing red circle centered on the affected joint to draw the athlete's attention to the mistake.[15]
- **Fading Movement Trails**: During slow-motion playback, the system displays an orange tracking path that traces the last 18 coordinate positions of the joint, helping players visualize their movement patterns.[15]
- **Real-Time Angle Arcs**: The platform draws dynamic angle arcs directly on the video frames. These arcs update in real time as the joints move, showing the measured angles alongside the target threshold targets.[15]
- **Auto-Pause and Scrubbing Hooks**: When a user selects a flagged issue, the player automatically pauses the video 2.5 seconds before the mistake occurs.[15] It then enables a

"Continue in slow-mo" button to play through the key movement sequence at $0.25\times$ speed.[15]

Biomechanical Coaching Cards

To make the system's feedback highly actionable, every flagged issue must be paired with a dedicated "Coaching Card".[15] These cards break down complex biomechanical data into plain-English feedback and structured training recommendations.[15]
The table below outlines the core structure of the Coaching Cards on cricketcoach.online.

Card Component	Development Data Source	Visual Presentation & Actionable Content
Biomechanical Pathology [15]	Rule engine trigger flag [15]	Simple, plain-English description of the technical error (e.g., "Front knee collapsing at landing").[15]
Tactical Context [15]	Physics engine and coordinate tracking [15]	Explains why the error matters, connecting the physical movement to performance loss or injury risks.[15]
Technical Adjustment Cue [15]	Athlete Profile metadata (Handedness/Role) [15]	High-impact verbal and physical cues to help the athlete adjust their form.[15]
Remedial Practice Drill [15]	Database of video drills [11]	Provides a specific practice drill, including video instructions, designed to correct the mechanical error.[15]

Algorithmic Athlete Profiling

To build long-term user engagement, the platform uses automated match-play data and training analysis to construct behavioral profiles for each athlete.[9]
The system categorizes players into distinct technical archetypes based on their performance metrics:

- **Classicist Wildcard**: Batters who maintain highly stable head positions ($\text{tilt} \leq 3^\circ$) and consistent, textbook vertical bat angles during defensive and drive strokes.[15]

- **Destroyer Aspirant**: Batters characterized by rapid hand speeds ($\text{velocity} \geq 12\ \mathrm{m/s}$), high backlift trajectories, and wide knee stances designed for maximum power.[15]
- **Hard Hitter Wildcard**: Batters who display high hip-rotation velocity and significant front-knee extension during impact, indicating aggressive, power-hitting mechanics.[36]
- **Accumulator Economist**: Bowlers who display highly repeatable delivery actions ($\text{variance} \leq 5\%$) and land at least 80% of their deliveries on a consistent line and length.[14]

Implementation Roadmap and Code Architecture for the Agent

To translate these requirements into an active development project on cricketcoach.online, the engineering team should adopt a clean, modular python-based microservices architecture.[15] The backend code is organized into dedicated files to ensure clear separation of concerns: app.py handles the API routes and task queues, video_processor.py manages frame extraction, pose_detector.py wraps the MediaPipe model, and batting_rules.py and bowling_rules.py define the biomechanical threshold parameters.[15]

Modular Backend Blueprint

The following Python codebase provides a complete, runnable base implementation for the core biomechanical and ball tracking components. This code defines the mathematical tracking framework, standardizes Joint angle calculations, and integrates the trajectory tracking logic [6]:

```python
import cv2
import numpy as np
import mediapipe as mp
from scipy.interpolate import PchipInterpolator
from ultralytics import YOLO

class CricketBiomechanicalAnalyzer:
    def __init__(self, yolo_model_path: str):
        # Initialize Google MediaPipe Pose engine [15, 24]
        self.mp_pose = mp.solutions.pose
        self.pose = self.mp_pose.Pose(
            static_image_mode=False
```

```
            model_complexity=1,
            enable_segmentation=False,
            min_detection_confidence=0.5
        )
        # Initialize custom ball tracking YOLO network [6, 31]
        self.ball_detector = YOLO(yolo_model_path)

    def calculate_joint_angle(self, p_start: np.ndarray, p_vertex: np.ndarray, p_end: np.ndarray) -> float:
        """
        Calculates the internal 2D/3D angle in degrees formed by three coordinate points.
        """
        u = p_start - p_vertex
        v = p_end - p_vertex

        cosine_angle = np.dot(u, v) / (np.linalg.norm(u) * np.linalg.norm(v) + 1e-6)
        angle = np.arccos(np.clip(cosine_angle, -1.0, 1.0))
        return float(np.degrees(angle))

    def evaluate_bowling_kinematics(self, pose_landmarks, width: int, height: int) -> dict:
        """
        Extracts key coordinates and evaluates bowler joint alignment parameters.
        """
        # Map MediaPipe landmark nodes to absolute pixel coordinates [15, 18]
        landmarks = {}
        target_nodes = {
            'left_hip': self.mp_pose.PoseLandmark.LEFT_HIP,
            'left_knee': self.mp_pose.PoseLandmark.LEFT_KNEE,
            'left_ankle': self.mp_pose.PoseLandmark.LEFT_ANKLE,
            'right_shoulder': self.mp_pose.PoseLandmark.RIGHT_SHOULDER,
            'right_elbow': self.mp_pose.PoseLandmark.RIGHT_ELBOW,
            'right_wrist': self.mp_pose.PoseLandmark.RIGHT_WRIST
        }

        for name, idx in target_nodes.items():
            lm = pose_landmarks.landmark[idx]
            if lm.visibility > 0.5:
                landmarks[name] = np.array([lm.x * width, lm.y * height])

        evaluation = {"status": "Incomplete Detection", "metrics": {}, "issues": []}

        # Verify mandatory joints are visible for bowling-side evaluation
        required_keys = ['left_hip', 'left_knee', 'left_ankle', 'right_shoulder', 'right_elbow', 'right_wrist']
```

```python
        if all(k in landmarks for k in required_keys):
            front_knee_angle = self.calculate_joint_angle(
                landmarks['left_hip'], landmarks['left_knee'], landmarks['left_ankle']
            )
            bowling_elbow_angle = self.calculate_joint_angle(
                landmarks['right_shoulder'], landmarks['right_elbow'], landmarks['right_wrist']
            )

            evaluation["metrics"]["front_knee_angle"] = front_knee_angle
            evaluation["metrics"]["bowling_elbow_angle"] = bowling_elbow_angle
            evaluation["status"] = "Success"

            # Run threshold evaluations against target physiological metrics
            if front_knee_angle < 155.0:
                evaluation["issues"].append({
                    "joint": "Front Knee",
                    "severity": "High",
                    "measurement": f"{front_knee_angle:.1f}°",
                    "pathology": "Knee collapses below 155°. This redirects ground forces directly to the spine, raising injury risks.",
                    "fix": "Focus on a firm front foot contact phase, keeping the landing knee locked."
                })

            if bowling_elbow_angle > 15.0:
                # Flag potential illegal throwing action based on joint angle
                evaluation["issues"].append({
                    "joint": "Bowling Elbow",
                    "severity": "Critical",
                    "measurement": f"{bowling_elbow_angle:.1f}°",
                    "pathology": "Elbow extension exceeds 15° limit, classifying the delivery as illegal.",
                    "fix": "Keep the arm extended throughout rotation, ensuring a high, vertical release arc."
                })

        return evaluation

    def track_ball_trajectory_pchip(self, video_path: str) -> list:
        """
       Runs YOLO object detection to locate the ball, applying PCHIP interpolation.[23, 31, 33]
       """
        cap = cv2.VideoCapture(video_path)
        frame_idx = 0
        raw_detections =
```

```python
    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break

        # Run YOLO ball object detection [31, 33]
        results = self.ball_detector(frame, verbose=False)
        detected = False

        for box in results.boxes:
            # Verify confidence scores and isolate target class indices [30, 33]
            if int(box.cls) == 0 and box.conf > 0.4:
                xyxy = box.xyxy.cpu().numpy()
                center_x = (xyxy + xyxy) / 2.0
                center_y = (xyxy + xyxy) / 2.0
                raw_detections.append((frame_idx, center_x, center_y))
                detected = True
                break

        if not detected:
            # Mark coordinate as missing for reconstruction
            raw_detections.append((frame_idx, None, None))

        frame_idx += 1
    cap.release()

    # Isolate valid frames and interpolate missing values
    detected_frames = [f for f, x, y in raw_detections if x is not None]
    detected_x = [x for f, x, y in raw_detections if x is not None]
    detected_y = [y for f, x, y in raw_detections if y is not None]

    if len(detected_frames) > 5:
        full_timeline = np.arange(frame_idx)
        # Apply PCHIP interpolation to reconstruct missing coordinates smoothly
        pchip_x = PchipInterpolator(detected_frames, detected_x)
        pchip_y = PchipInterpolator(detected_frames, detected_y)

        smoothed_trajectory = [
            (int(f), float(pchip_x(f)), float(pchip_y(f))) for f in full_timeline
        ]
        return smoothed_trajectory
```

```
    return [(f, x, y) for f, x, y in raw_detections]
```

This software blueprint provides the engineering team with the core algorithms required to build out high-performance, synchronized multi-camera analysis capabilities and advanced ball-tracking features for cricketcoach.online.[6]

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